Zhang 2019 follicular lymphoma — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

4/4 samples (≥100 cells) intermix on one continuous manifold with no per-patient islands (iLISI 1.00→2.26 of 4 at Harmony θ=10; same-patient nbr frac 0.60; 1 kNN component, 0 detached). Strongly clonal FL batch integrates into one body.

Gate 2 — module geography → NO-GO

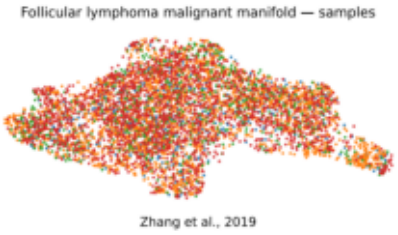
Embryonic programs are at the control-pool noise floor (means Gast +0.012 / pSom +0.006 / Crest +0.009, 20–40× below real-pattern datasets). No Gast/pSom hub (joint-high enrichment 1.0). Gast–pSom r and Crest pseudotime gradient sit INSIDE the random-module 95% null band (48th / 72nd null percentile) — no embryonic-specific signal. Pattern absent.

Gate 3 — cross-repression → NA

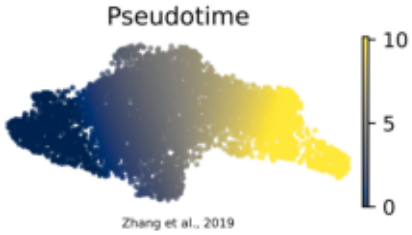
Not reached — Gate 2 NO-GO halts the gate chain. No bifurcation (curve: 2 tips, 0 forks) and no hub, so there is no branch pair to test for cross-repression.

STEP FIGURES

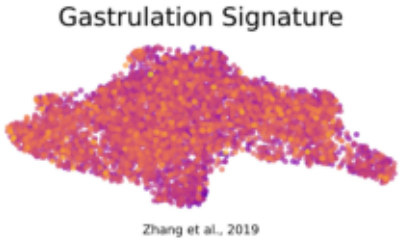
Manifold — samples



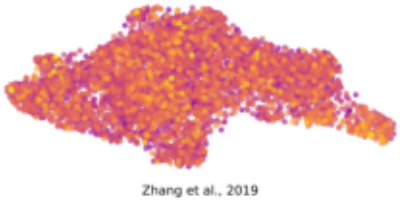
Pseudotime



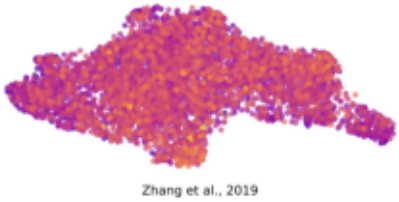
Gastrulation



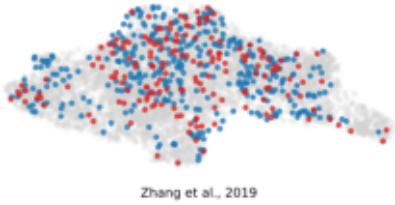
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.

Follicular lymphoma (Zhang et al., 2019), 10x — 2 patients (FL1018, FL2001) / 7,002 cells; malignant only: 5,645 B cells across 4 samples (all clear the ≥100 floor); whole malignant manifold (no lobe scoping). Pattern absent — Gate 2 NO-GO.

Verdict.

Pattern does NOT reproduce — Gate 1 GO, Gate 2 NO-GO, Gate 3 NA. The FL malignant manifold is one integrated continuum, but the gastrulation/somitogenesis/neural-crest programs are absent: means at the control-pool floor and, decisively, Gast–pSom co-variation and the Crest pseudotime gradient are indistinguishable from random gene sets of equal size (48th / 72nd null percentile). No Gast/pSom hub (enrichment 1.0). This corrects 5 prior attempts that called Gate 2 GO by reading score_genes shared-pool artifacts (and, in -4, a circularly-scoped 196-cell cycling lobe that flipped Gast–pSom to +0.33). EMT-TFs ZEB2 (28%) · ZEB1 (12%), both terminus-loaded; SNAI/TWIST/FOXC/PRRX below 3% detection.

Parameter choices

Min cells/sample	≥100 floor (4/4; gap 2.74 not a cliff) → 5,645 cells	**Scope**	whole manifold (no scoping; verified robust across scopes)
Root node	tip 2 = Gast/pSom-high, Crest-low	**Integration**	Harmony θ=10; PCA20; spread=1.3
Trajectory	scFates curve N=20, λ=0.3, μ=100 (fold-free, 2 tips / 0 forks)	**Decisive test**	200-draw random-gene-set null (same sizes/pool/smoothing)
Gate readout	G1 GO / G2 NO-GO / G3 NA (module means .012/.006/.009)	**EMT-TFs**	ZEB2 (28%) · ZEB1 (12%), both terminus-loaded; rest below 3% detection